

Lesson 11 Homework - Calculus

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We want to find the first and second order derivatives of

$$r = \frac{(\theta - 1)(\theta^2 + \theta + 1)}{\theta^3}$$

We can see that the numerator is the factorization of difference of cubes, hence we have

$$r = \frac{\theta^3 - 1}{\theta} = 1 + \frac{1}{\theta^3} = 1 + \theta^{-3}$$

Then differentiating using the power rule, we have

$$\frac{dr}{dt} = 1 - 3\theta^{-4} = 1 - \frac{3}{\theta^4}$$

Then the second order derivative is

$$\frac{d^2r}{dt^2} = \frac{d}{dt}(1 - 3\theta^{-4}) = 1 + \frac{12}{\theta^5}$$

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